Systematic Drug Repurposing Screen Identifies Multi-Cascade Therapeutic Targets for COL11A1-Associated Marshall Syndrome

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Date: March 2026
Version: 1.0 (initial preprint)

Abstract

Background: Marshall syndrome (OMIM #154780) is a rare autosomal dominant connective tissue disorder caused by mutations in *COL11A1* that produce a dominant-negative type XI collagen protein. No disease-modifying therapy exists, and no systematic drug repurposing analysis has previously been conducted for any type XI collagenopathy. Recent characterization of eight self-amplifying destructive cascades downstream of defective collagen XI provides a mechanistic framework for identifying pharmacological intervention points.

Methods: Eight self-amplifying cascades were systematically identified from published literature on disc degeneration, cellular senescence, ferroptosis, complement biology, and extracellular matrix remodeling, and mapped to the COL11A1 dominant-negative pathomechanism. For each cascade, existing FDA-approved or commercially available compounds were screened for mechanism-of-action overlap with cascade nodes. Candidates were ranked by a composite score incorporating evidence quality, safety profile, number of cascades addressed, pediatric applicability, and practical availability. Network connectivity analysis was performed on the eight-cascade system to identify high-leverage intervention points.

Results: Thirty-four drug and supplement candidates were identified across seven of eight cascades. Network analysis revealed that the Senescence/SASP cascade (Loop 3) functions as the principal hub node with eight inter-cascade connections, followed by the MMP/ADAMTS cascade (Loop 1, five connections) and the Ferroptosis cascade (Loop 2, five connections). The top-ranked candidates were N-acetylcysteine (NAC; direct GPX4 substrate, anti-ferroptotic),

metformin (AMPK activator; IL-6 reduction of 58%, MMP-3 reduction of 64% in clinical data; three cascades), and sub-antimicrobial doxycycline (20 mg; the only FDA-approved MMP inhibitor). A three-tiered combination strategy covering supplements, prescription medications, and specialist-level interventions was constructed that theoretically addresses seven of eight cascades, with the Connexin-43 death wave cascade remaining the sole untargeted pathway due to the absence of a selective, safe hemichannel inhibitor. Ten drugs were prioritized for potential N-of-1 trial evaluation.

Conclusions: This is, to our knowledge, the first systematic drug repurposing screen for any type XI collagenopathy. The cascade-based screening approach identifies actionable pharmacological targets using existing approved drugs. Clinical validation through structured N-of-1 trials with biomarker endpoints is the logical next step. The dominant-negative mechanism of COL11A1 mutations additionally creates a favorable landscape for future allele-specific gene silencing therapies, which have been proven effective in four sister collagen disorders.

1. Introduction

1.1 Marshall Syndrome and COL11A1

Marshall syndrome (OMIM #154780) is a rare autosomal dominant connective tissue disorder caused by heterozygous mutations in the *COL11A1* gene encoding the alpha-1 chain of type XI collagen [1, 2]. Type XI collagen is a quantitatively minor but functionally critical fibrillar collagen that regulates the diameter and spacing of type II collagen fibrils in cartilage, vitreous humor, the tectorial membrane of the inner ear, and the nucleus pulposus of intervertebral discs [3]. The clinical phenotype includes sensorineural hearing loss, high myopia with risk of retinal detachment, midface hypoplasia, early-onset osteoarthritis, and progressive spinal degeneration with disc disease and vertebral anomalies [1, 4].

The majority of pathogenic *COL11A1* variants in Marshall syndrome are splice-site mutations that result in in-frame exon skipping, producing a shortened but translatable protein [2, 5]. This truncated alpha-1(XI) chain is incorporated into collagen XI trimers, where it disrupts the triple-helical structure from within—a **dominant-negative** mechanism. Critically, *COL11A1* is haplosufficient: individuals carrying null alleles that produce no protein from one copy are clinically unaffected [5, 6]. This establishes that a single functional copy of *COL11A1* is sufficient for normal collagen XI function, and that the disease phenotype arises not from collagen deficiency but from active poisoning of the collagen network by the mutant protein.

1.2 The Unmet Therapeutic Need

No disease-modifying therapy exists for Marshall syndrome or any type XI collagenopathy. Current management is limited to symptomatic treatment: hearing aids or cochlear implants for hearing loss, prophylactic retinal procedures for detachment risk, pain management for spinal

and joint symptoms, and surgical intervention for advanced structural damage [4]. No clinical trial has ever been conducted specifically for Marshall syndrome, and no drug repurposing analysis targeting the downstream molecular cascades has been published for any COL11A1-associated condition.

A 2025 network pharmacology analysis of Stickler syndrome published in *PLOS ONE* identified candidate molecules through gene-network approaches but concluded that “the complex genetic maze remains poorly understood, limiting targeted therapeutics” [7]. That analysis did not address the specific self-amplifying destructive cascades that have since been characterized as drivers of tissue damage in COL11A1-dominant disease.

1.3 The Cascade Model

Recent synthesis of published literature from disc biology, cellular senescence, ferroptosis, complement immunology, and extracellular matrix remodeling has identified eight self-amplifying destructive cascades that operate downstream of the COL11A1 dominant-negative mechanism [8]. Unlike the traditional “weak collagen” model, which treats Marshall syndrome as a passive structural deficiency, the cascade model describes an active, progressive process in which the body’s own biological responses to defective collagen create self-reinforcing feedback loops that accelerate tissue destruction over time.

This cascade framework provides, for the first time, a structured set of molecular targets amenable to pharmacological intervention. The present study exploits this framework to conduct a systematic drug repurposing screen: identifying existing, approved drugs whose mechanisms of action intersect with nodes within these eight cascades.

2. Methods

2.1 Cascade Identification and Characterization

Eight self-amplifying destructive cascades were identified through systematic review of peer-reviewed literature in the following domains: intervertebral disc degeneration, cartilage biology, cellular senescence, ferroptosis, complement immunology, gap junction biology, extracellular matrix remodeling, and the gut-disc axis. Each cascade was required to meet three inclusion criteria: (i) a demonstrated self-amplifying (positive feedback) mechanism in published literature; (ii) a plausible mechanistic connection to COL11A1-mediated pathology; and (iii) at least one identified molecular node amenable to pharmacological intervention.

The eight cascades and their principal molecular nodes were:

1. **MMP/ADAMTS Fragment Cascade** — MMP-1, MMP-3, MMP-8, MMP-9, MMP-13, ADAMTS-5, NF-kB, DDR2

2. **Ferroptosis Pathway** — GPX4, labile iron pool, ALOX5, lipid peroxidation, PROMININ2/CD63 exosomes
3. **Senescence/SASP Cascade** — p16/p21, mTOR, AMPK, NF-kB, IL-6, SASP components, senescent cell survival networks (SRC, PI3K, BCL-2, serpines)
4. **Connexin-43 Death Waves** — Cx43 hemichannels, gap junctions, ATP/calcium propagation
5. **Endplate Calcification Cycle** — HIF-2alpha, Matrix Gla Protein (MGP), RUNX2, nutrient diffusion
6. **Complement Sump** — C3, C5, C5a receptor, membrane attack complex (MAC, C5b-9)
7. **Lactate-Senescence Loop** — anaerobic glycolysis, lactate accumulation, mitochondrial dysfunction, mTOR/AMPK
8. **Gut-Disc Triangle** — gut barrier integrity (tight junctions, zonulin), LPS translocation, TLR4/NF-kB, microbiome composition, short-chain fatty acids

2.2 Drug Candidate Identification

For each cascade, existing compounds were identified through mechanism-of-action matching against the cascade's principal molecular nodes. The search space included: (i) FDA-approved drugs, regardless of original indication; (ii) drugs approved by other regulatory agencies (EMA, TGA) but not FDA-approved; (iii) commercially available dietary supplements with published pharmacological data; and (iv) investigational drugs with completed Phase 1 or Phase 2 human safety data.

Candidate identification was performed through PubMed literature search (terms: each molecular node + "inhibitor" OR "modulator" OR "drug repurposing"), examination of FDA drug labels for mechanism-of-action statements, review of ClinicalTrials.gov for active or completed trials, and cross-referencing with the DrugBank and ChEMBL databases.

2.3 Ranking Methodology

Candidates were ranked by a composite score incorporating five weighted domains:

Domain	Weight	Scoring Criteria
Evidence quality	30%	Strong = human clinical data or FDA-approved for MoA; Moderate = preclinical data in relevant tissue; Weak = mechanistic rationale only
Safety profile	25%	Good = decades of clinical use, including pediatric; Moderate = established safety but significant monitoring required; Caution = significant adverse effect risk or limited long-term data

Domain	Weight	Scoring Criteria
Cascade coverage	20%	Number of distinct cascades addressed by the compound
Pediatric applicability	15%	Existing pediatric dosing data, pediatric safety record, growth-related contraindications
Practical availability	10%	OTC supplement > generic Rx > specialist Rx > IV-only > investigational

2.4 Network Connectivity Analysis

Inter-cascade connections were mapped by reviewing published literature for demonstrated or mechanistically plausible feed-forward and feedback relationships between each pair of cascades. A directed graph was constructed with cascades as nodes and feeding relationships as edges. For each node, in-degree (number of cascades feeding it), out-degree (number of cascades it feeds), and total degree (sum of in-degree and out-degree) were calculated. Nodes were prioritized by total degree, with ties broken by in-degree (higher in-degree indicates greater susceptibility to upstream disruption).

3. Results

3.1 The Eight Cascades and Their Interconnections

All eight identified cascades met the three inclusion criteria (self-amplifying mechanism, connection to COL11A1 pathology, and at least one targetable node). The central mechanism linking the COL11A1 mutation to cascade activation is as follows: defective collagen XI, produced by the dominant-negative mutant protein, is incorporated into the extracellular matrix, where it disorganizes fibril architecture [3]. Enzymatic degradation of the disorganized matrix generates collagen fragments that act as danger-associated molecular patterns (DAMPs), activating Wnt/AKT/GSK3-beta signaling, NF-kB inflammatory pathways, and stress-induced premature senescence (SIPS) in resident cells [8, 9, 10]. These activated pathways drive the eight cascades described below.

3.1.1 Cascade Descriptions

Cascade 1: MMP/ADAMTS Fragment Cascade. Collagen fragments generated by the initial structural defect activate DDR2 receptors and NF-kB, upregulating matrix metalloproteinases (MMP-1, -3, -8, -9, -13) and ADAMTS aggrecanases [9, 10]. These enzymes degrade additional collagen and proteoglycans, producing more fragments and creating a feed-forward destruction loop. This cascade is the primary enzymatic effector of tissue destruction in COL11A1 disease.

Cascade 2: Ferroptosis Pathway. Iron-dependent lipid peroxidation drives a specific form of regulated cell death (ferroptosis) in disc and cartilage tissue [11, 12]. Labile iron accumulates from senescent cells (which release 30-fold more iron than healthy cells), complement MAC-pore formation, and dysregulated ferritinophagy [12, 13]. GPX4, a selenoprotein, is the sole enzymatic defense against ferroptotic death [11]. Iron-loaded exosomes bearing PROMININ2/CD63 surface markers propagate ferroptotic signals to neighboring cells [8].

Cascade 3: Senescence/SASP Cascade. Collagen fragments and mechanical stress induce stress-induced premature senescence (SIPS) in disc cells [14, 15]. Senescent cells secrete the senescence-associated secretory phenotype (SASP)—a cocktail of IL-6, IL-8, MMP-3, MMP-13, and other inflammatory mediators that degrades the surrounding matrix and converts neighboring cells to senescence via paracrine signaling [14, 16]. Exosomes carrying miR-27a-3p and lipocalin-2 (LCN2) further propagate senescence to distant cells [8].

Cascade 4: Connexin-43 Death Waves. Gap junctions formed by connexin-43 (Cx43) physically connect disc and cartilage cells. When one cell enters crisis, death signals (ATP, calcium) propagate through Cx43 channels at approximately 5.5 micrometers per minute, triggering cascading cell death across connected tissue [8, 17]. This cascade amplifies the damage from every other cascade by propagating it to neighboring cells.

Cascade 5: Endplate Calcification Cycle. Vertebral endplates—the primary nutrient transport route into the avascular disc—undergo progressive calcification driven by the hypoxia-responsive transcription factor HIF-2alpha and RUNX2 [8, 18, 19]. Calcification reduces nutrient flow, which stresses disc cells, which release more calcification-promoting signals, creating a self-sealing nutrient deprivation loop.

Cascade 6: Complement Sump. Complement proteins (C3, C5) accumulate in the avascular disc interior, where they cannot be cleared by blood flow [8, 20]. Collagen fragments activate the complement cascade, and the membrane attack complex (MAC, C5b-9) punches pores in disc cell membranes, allowing iron influx that feeds ferroptosis (Cascade 2) [13, 20, 21].

Cascade 7: Lactate-Senescence Loop. Disc cells in the low-oxygen, nutrient-deprived disc environment rely on anaerobic glycolysis, producing excess lactate [8, 22]. High lactate concentrations promote cellular senescence (Cascade 3), and senescent cells have dysfunctional mitochondria that produce even more lactate, creating a metabolic trap [8].

Cascade 8: Gut-Disc Triangle. COL11A1 is expressed in the intestinal submucosa, meaning the gut barrier is structurally compromised in Marshall syndrome [8, 23]. Increased gut permeability allows bacterial endotoxin (LPS) translocation, which activates TLR4/NF-kB inflammatory cascades systemically and in the disc [23, 24]. SASP-driven systemic inflammation (from Cascade 3) further disrupts the gut microbiome, creating a three-way feedback loop between senescent cells, gut bacteria, and ferroptotic cell death [8].

3.1.2 Network Connectivity Analysis

The eight cascades form a densely interconnected network. The connectivity analysis is summarized in Table 1.

Cascade	Feeds (Out-Degree)	Fed By (In-Degree)	Total Connections	Priority Rank
3. Senescence/SASP	1, 7, 8	1, 2, 4, 5, 7	8	**HIGHEST**
1. MMP/ADAMTS	3, 8	3, 6, 7	5	**HIGH**
2. Ferroptosis	1, 3	4, 5, 8	5	**HIGH**
8. Gut-Disc Triangle	1, 2, 3	3	4	MODERATE
7. Lactate-Senescence	3, 1	3	3	MODERATE
5. Endplate Calcification	2, 3	—	2	MODERATE
4. Cx43 Death Waves	2, 3	—	2	LOW (no drug target)
6. Complement	1, 3	—	2	LOW

Table 1. Network connectivity analysis of the eight self-amplifying cascades in COL11A1-associated Marshall syndrome. Total connections = out-degree + in-degree. Priority reflects both connectivity and druggability.

Key finding: The Senescence/SASP cascade (Loop 3) is the principal hub node in the cascade network, with eight total connections (three outgoing, five incoming). It feeds three other cascades and is fed by five, making it both the largest amplifier and the most accessible leverage point for pharmacological intervention. Disrupting Cascade 3 would theoretically reduce input to Cascades 1, 7, and 8 while being partially starved by disruption of any of Cascades 1, 2, 4, 5, or 7.

3.2 Drug Screen Results by Cascade

Thirty-four distinct compounds were identified across the eight cascades. Table 2 presents the complete screen results organized by cascade.

Cascade 1: MMP/ADAMTS Fragment Cascade (5 candidates)

Drug	Status	Mechanism	Evidence	Safety	Cascades Hit
Sub-antimicrobial doxycycline (20 mg)	FDA-approved	Only FDA-approved MMP inhibitor; inhibits MMP-1, -3, -8, -9, -13 at sub-antimicrobial dose without antibiotic activity or gut microbiome disruption [25]	Strong	Moderate	MMP
Simvastatin/Atorvastatin	FDA-approved	Prevents MMP-mediated collagen breakdown via Rho/ROCK inhibition; suppresses NF-kB and Wnt/beta-catenin [26]	Moderate	Good	MMP, NF-kB, Wnt
Colchicine	FDA-approved	Suppresses NF-kB and NLRP3 inflammasome; reduces neutrophil MMP release; pediatric dosing (age 4+) [27]	Strong	Moderate	NF-kB, NLRP3, Complement
Diacerein	EMA-approved	IL-1 inhibitor reducing downstream MMP expression; OA-approved in Europe [28]	Moderate	Moderate	MMP, IL-1

Drug	Status	Mechanism	Evidence	Safety	Cascades Hit
Curcumin (Theracurmin/CMG)	Supplement	Blocks MMP-13 transcription via NF-kB inhibition; inhibits ADAMTS-5; requires bioavailable formulation (27–40x absorption) [29]	Moderate	Good	MMP, NF-kB

Cascade 2: Ferroptosis Pathway (6 candidates)

Drug	Status	Mechanism	Evidence	Safety	Cascades Hit
N-Acetylcysteine (NAC)	FDA-approved/OTC	2025 research confirmed NAC directly serves as a GPX4 substrate, not merely a glutathione precursor; direct anti-ferroptotic agent [30, 11]	Strong	Good	Ferroptosis, ER stress
Selenium (selenomethionine)	Supplement	Essential GPX4 cofactor; GPX4 is a selenoprotein that cannot function without selenium [31, 11]	Strong	Good	Ferroptosis
Vitamin E (tocotrienols)	Supplement	Lipid radical-trapping antioxidant cooperating with GPX4; tocotrienols 40–60x more potent	Moderate	Good	Ferroptosis

Drug	Status	Mechanism	Evidence	Safety	Cascades Hit
		than tocopherols for membrane radical scavenging [12]			
Deferiprone	FDA-approved	Oral iron chelator crossing blood-brain barrier; reduces labile iron pool fueling Fenton chemistry [32]	Moderate	Caution	Ferroptosis
Zileuton	FDA-approved	ALOX5 inhibitor blocking ferroptotic lipid peroxide generation at enzymatic source [12]	Moderate	Moderate	Ferroptosis, Inflammation
Lactoferrin	Supplement	Iron-binding glycoprotein sequestering labile iron; strengthens gut barrier integrity [24]	Moderate	Good	Ferroptosis, Gut barrier

Cascade 3: Senescence/SASP (5 candidates)

Drug	Status	Mechanism	Evidence	Safety	Cascades Hit
Rapamycin/Sirolimus	FDA-approved	mTOR inhibitor; senomorphic (suppresses SASP); reduces glycolytic flux. Contraindicated during growth [16]	Strong	Caution	SASP, Lactate, mTOR

Drug	Status	Mechanism	Evidence	Safety	Cascades Hit
Metformin	FDA-approved	AMPK activation suppresses NF-kB and SASP; IL-6 reduced 58%, MMP-3 reduced 64% in TAME trial; pediatric use established [33]	Strong	Good	MMP, SASP, Lactate
Quercetin	Supplement	Senolytic at high intermittent doses; inhibits MMP-9 and NF-kB [34, 35]	Moderate	Good	MMP, Senolytic
Fisetin	Supplement	More potent senolytic than quercetin; significant senescent cell clearance in aged mice [36]	Moderate	Good	Senolytic
Dasatinib + Quercetin (D+Q)	Investigational	Most-studied senolytic combination; dasatinib targets SRC kinases, quercetin inhibits PI3K/BCL-2; Phase 1/2 trials in IPF, DKD, Alzheimer's [34, 37]	Moderate	Moderate	Senolytic

Cascade 4: Connexin-43 Death Waves (2 candidates)

Drug	Status	Mechanism	Evidence	Safety	Cascades Hit
Carbenoxolone	Historical	Non-selective gap junction blocker; significant mineralocorticoid side	Moderate	Caution	Cx43

Drug	Status	Mechanism	Evidence	Safety	Cascades Hit
		effects (hypertension, hypokalemia) limit use [17]			
Tonabersat	Investigational	Selective Cx43 hemichannel inhibitor preserving gap junction communication; Phase 2 human safety data from migraine trials; not approved [17, 38]	Moderate	Moderate	Cx43

Cascade 4 represents the largest unmet pharmacological need. No approved drug safely and selectively blocks Cx43 hemichannels without affecting cardiac gap junctions. This cascade remains the sole untargeted pathway in the repurposing screen.

Cascade 5: Endplate Calcification (3 candidates)

Drug	Status	Mechanism	Evidence	Safety	Cascades Hit
Vitamin K2 (MK-7)	Supplement	Activates Matrix Gla Protein (MGP), the body's most powerful natural inhibitor of soft-tissue calcification; monitorable via dp-ucMGP [18, 39]	Strong	Good	Calcification
Alendronate	FDA-approved	Bisphosphonate retarding endplate thickening; reduced MMP-1, -3, -13 in disc tissue in animal models; pediatric OI experience [40]	Moderate	Moderate	Calcification, MMP
Belzutifan	FDA-approved	HIF-2alpha inhibitor; side effect profile (anemia, hypoxia) makes it impractical for chronic non-cancer use [19]	Moderate	Caution	Calcification

Cascade 6: Complement (3 candidates)

Drug	Status	Mechanism	Evidence	Safety	Cascades Hit
Hydroxychloroquine	FDA-approved	Complement modulation; decades of connective tissue disease experience; pediatric dosing available; requires ophthalmologic monitoring [20]	Moderate	Moderate	Complement, Inflammation
Avacopan	FDA-approved	Oral C5a receptor inhibitor; more targeted than broad complement inhibition; limited pediatric data [21]	Moderate	Moderate	Complement
Eculizumab	FDA-approved	Anti-C5 monoclonal antibody; effective but impractical: cost >$500,000/year, IV-only, meningococcal infection risk [20]	Strong	Caution	Complement

Cascade 7: Lactate-Senescence Loop (2 candidates, shared with Cascade 3)

Metformin and rapamycin, described under Cascade 3, also address Cascade 7 through AMPK-mediated metabolic shifting (metformin) and mTOR-dependent glycolytic flux reduction (rapamycin) [22, 33].

Cascade 8: Gut-Disc Triangle (4 candidates)

Drug	Status	Mechanism	Evidence	Safety	Cascades Hit
Probiotics (Lactobacillus/Bifidobacterium)	Supplement	26-RCT meta-analysis: reduced zonulin and circulating LPS;	Strong	Good	Gut barrier

Drug	Status	Mechanism	Evidence	Safety	Cascades Hit
		increased butyrate production suppressing NF-kB [24, 41]			
Sodium butyrate/Tributyrin	Supplement	HDAC inhibitor supporting tight junction integrity; senomorphic properties; colonocyte energy substrate [24]	Moderate	Good	Gut barrier, SASP
Omega-3 (EPA/DHA)	Supplement	Gut barrier support; resolvin precursor (active inflammation resolution); shifts membrane composition away from pro-ferroptotic lipids [12, 24]	Moderate	Good	Ferroptosis, Gut barrier, Inflammatio
Lactoferrin	Supplement	Iron binding in gut lumen; gut barrier strengthening via LPS neutralization and tight junction protein upregulation [24]	Moderate	Good	Ferroptosis, Gut barrier

Additional Multi-Cascade Compounds (7 candidates)

Drug	Status	Mechanism	Evidence	Safety
Losartan	FDA-approved	TGF-beta modulation via AT1 receptor blockade; 8-year efficacy data in Marfan syndrome; pediatric dosing established [42]	Strong	Good
4-Phenylbutyrate (4-PBA)	FDA-approved	Chemical chaperone reducing ER stress from misfolded COL11A1 protein; FDA-approved for urea cycle disorders [43]	Moderate	Good
Pentoxifylline	FDA-approved	TNF-alpha inhibitor with anti-fibrotic and hemorrheologic properties; may improve endplate perfusion [10]	Moderate	Good
Low-Dose Naltrexone (LDN)	Off-label	TLR4 antagonism and microglial suppression at low doses (1–4.5 mg); addresses central sensitization and neuroinflammation [44]	Moderate	Good
Pirfenidone	FDA-approved	Broad anti-fibrotic (TGF-beta, TNF-alpha, PDGF); relevant to late-stage fibrotic transformation; significant GI side effects [10]	Moderate	Moderate
Anakinra	FDA-approved	IL-1 receptor antagonist blocking SASP component IL-1beta; pediatric safety data (CAPS); daily SC injection [10]	Strong	Moderate
Resveratrol	Supplement	SIRT1 activator inhibiting MMP-3 in nucleus pulposus cells; poor bioavailability without micronization [10]	Moderate	Good

3.3 Top 10 Ranking

Table 3 presents the top 10 candidates ranked by the composite score (Section 2.3).

Rank	Drug	Cascades Addressed	Evidence	Safety	Availability
1	**N-Acetylcysteine (NAC)**	Ferroptosis, ER stress	Strong (2025 GPX4 substrate data) [30]	Good	OTC + Rx
2	**Metformin**	MMP, SASP, Lactate	Strong (IL-6 −58%, MMP-3 −64%) [33]	Good	Rx (generic)
3	**Sub-antimicrobial doxycycline (20 mg)**	MMP (direct inhibition)	Strong (only FDA-approved MMP inhibitor) [25]	Moderate	Rx
4	**Vitamin K2 (MK-7)**	Endplate calcification	Strong (MGP activation) [39]	Good	Supplement
5	**Selenium**	Ferroptosis	Strong (essential GPX4 cofactor) [31]	Good	Supplement
6	**Curcumin (Theracurmin)**	MMP, NF-kB	Moderate [29]	Good	Supplement
7	**Probiotics (multi-strain)**	Gut barrier	Strong (26-RCT meta-analysis) [41]	Good	Supplement
8	**Quercetin**	MMP, Senolytic	Moderate [34, 35]	Good	Supplement
9	**Colchicine**	NF-kB, NLRP3, Complement	Strong [27]	Moderate	Rx (generic)
10	**Omega-3 (EPA/DHA)**	Ferroptosis, Gut barrier, Inflammation	Moderate [12, 24]	Good	Supplement

Table 3. Top 10 drug candidates ranked by composite score (evidence quality 30%, safety 25%, cascade coverage 20%, pediatric applicability 15%, availability 10%). None have been clinically tested for Marshall syndrome.

3.4 Combination Strategy

A three-tiered combination strategy was constructed to maximize cascade coverage while respecting safety and availability constraints, escalating from the most accessible interventions to those requiring specialist oversight.

Tier 1: Supplements (OTC, physician-informed). NAC + Selenium + Vitamin K2 (MK-7) + Vitamin E (tocotrienols) + Omega-3 (EPA/DHA) + Probiotics (multi-strain) + Curcumin (Theracurmin/CMG formulation). Cascades addressed: Ferroptosis (2), Endplate Calcification (5), Gut-Disc Triangle (8), MMP/NF-kB (partial 1).

Tier 2: Prescription medications (physician-directed). Sub-antimicrobial doxycycline (20 mg) + Metformin + Losartan + Low-Dose Naltrexone. Additional cascades addressed: MMP direct inhibition (1), SASP (3), Lactate-Senescence (7), TGF-beta pathway, neuroinflammation.

Tier 3: Specialist-level interventions (rheumatology/immunology oversight). Colchicine, hydroxychloroquine, or rapamycin (post-growth only). Additional cascades addressed: NLRP3 inflammasome, Complement (6), mTOR/SASP (3).

> **Cascade coverage summary.** The full three-tier strategy theoretically addresses **7 of 8 cascades** (all except Connexin-43 death waves). This represents near-complete pharmacological coverage of the identified cascade network using exclusively existing, approved drugs and commercially available supplements. The Cx43 cascade remains the sole untargeted pathway; tonabersat (Cx43 hemichannel inhibitor with Phase 2 human safety data) may eventually fill this gap if development continues.

3.5 Cascade Coverage Network Analysis

To characterize the overall pharmacological coverage of the cascade network, the proportion of cascades targetable by existing drugs was assessed. Seven of eight cascades (87.5%) have at least one identified drug candidate with published mechanism-of-action data supporting intervention at a cascade node. Six of eight cascades (75.0%) have at least one FDA-approved drug candidate. Five of eight cascades (62.5%) have at least one commercially available OTC supplement candidate.

The multi-cascade drugs—particularly metformin (Cascades 1, 3, 7), omega-3 (Cascades 2, 8), and NAC (Cascade 2 plus ER stress)—are especially noteworthy for their ability to simultaneously address multiple cascade nodes with a single compound, reducing polypharmacy burden.

4. Discussion

4.1 Novelty and Significance

To our knowledge, this is the first systematic drug repurposing screen conducted for any type XI collagenopathy. Previous work on Stickler syndrome pharmacology was limited to a single in-silico network pharmacology analysis [7] that did not address the specific self-amplifying cascades now understood to drive tissue destruction in COL11A1-dominant disease. The present analysis is the first to: (i) systematically map the eight-cascade network architecture, (ii) identify its hub node (Senescence/SASP, Cascade 3), (iii) screen existing drugs against individual cascade nodes, (iv) rank candidates by a multi-domain composite score, and (v) construct a tiered combination strategy achieving near-complete cascade coverage.

The finding that 7 of 8 cascades are targetable with existing, approved drugs or commercially available supplements is clinically significant. It suggests that disease modification in Marshall syndrome may not need to await novel drug development or gene therapy, but could potentially be initiated with appropriately validated off-label use of existing compounds—subject to rigorous clinical validation.

4.2 The Hub Node: Senescence/SASP as Therapeutic Priority

The network connectivity analysis identifies the Senescence/SASP cascade (Loop 3) as the principal hub with eight total connections. This finding has direct therapeutic implications. Drugs targeting Cascade 3—metformin (senomorphic), quercetin and fisetin (senolytic), rapamycin (senomorphic)—would be predicted to have the greatest cascade-wide impact per drug. The clinical data supporting metformin are particularly compelling: the TAME trial biomarker substudy demonstrated a 58% reduction in IL-6 and 64% reduction in MMP-3, two key SASP components that also drive Cascade 1 [33]. These reductions were observed at standard antidiabetic doses in a well-tolerated, generic medication with decades of pediatric safety data for insulin resistance.

The emergence of pharmaceutical senolytics (dasatinib + quercetin, fisetin) in clinical trials for age-related diseases [34, 36, 37] creates additional opportunities. If the senolytic hypothesis is validated in human trials—that intermittent clearance of senescent cells can reduce SASP burden and slow tissue degeneration—the implications for Marshall syndrome would be substantial, given that premature senescence is a demonstrated consequence of COL11A1-mediated collagen fragment signaling [14, 15].

4.3 The Connexin-43 Gap

Cascade 4 (Cx43 death waves) remains the sole untargeted pathway in this screen. No approved drug safely and selectively blocks Cx43 hemichannels without also affecting cardiac gap

junctions, which are essential for normal cardiac conduction [17]. Carbenoxolone is non-selective and carries prohibitive mineralocorticoid side effects. Tonabersat, which selectively inhibits Cx43 hemichannels while preserving gap junction communication, has completed Phase 2 trials for migraine and has human safety data [38], but is not approved for any indication and is not currently in development for musculoskeletal applications. The Cx43 cascade thus represents the highest-priority target for novel drug development in the context of COL11A1-associated disease.

4.4 The Dominant-Negative Advantage for Gene Therapy

While this screen focuses on pharmacological intervention, the dominant-negative mechanism of COL11A1 mutations creates a uniquely favorable landscape for definitive gene therapy. Because *COL11A1* is haplosufficient—one functional copy produces sufficient collagen XI for normal function—the therapeutic goal is not gene replacement but allele-specific silencing: suppressing expression of the mutant allele while leaving the normal allele intact [5, 6]. This approach has been proven effective in four sister collagen disorders: COL1A1 and COL1A2 (osteogenesis imperfecta; allele-specific siRNA) [45], COL6A1 (Ullrich dystrophy/Bethlem myopathy; allele-specific CRISPR, >40% silencing with matrix restoration) [46], and COL6A3 (collagen VI myopathy; gapmer ASOs) [46]. The molecular logic is identical across all dominant-negative collagen disorders: the mutation itself provides a distinctive sequence handle for allele-specific targeting.

The cascade-breaking pharmacological strategy described here is complementary to, not competitive with, gene therapy. Even after successful allele-specific silencing, established cascades may continue to operate autonomously for a period, sustained by accumulated senescent cells, iron deposits, and calcified endplates. A combined approach—gene therapy to halt production of the toxic protein, plus cascade-breaking drugs to disrupt the self-amplifying loops already in motion—may ultimately prove more effective than either strategy alone.

4.5 Implications for N-of-1 Trials

The extreme rarity of Marshall syndrome (estimated fewer than 1 in 100,000 live births) makes traditional randomized controlled trials impractical. The N-of-1 trial design, in which a single patient serves as their own control through structured crossover periods with biomarker endpoints, is the most feasible path to clinical validation [47]. The cascade model provides specific, measurable biomarker endpoints for each targeted cascade:

- **Cascade 1 (MMP):** serum MMP-3, urinary CTX-II (cartilage degradation)
- **Cascade 2 (Ferroptosis):** serum ferritin, transferrin saturation, malondialdehyde (MDA)
- **Cascade 3 (Senescence):** serum IL-6, $p16^{INK4a}$ expression in circulating T cells

- **Cascade 5 (Calcification):** dephospho-uncarboxylated MGP (dp-ucMGP)
- **Cascade 6 (Complement):** C3a, C5a, sC5b-9
- **Cascade 8 (Gut):** serum zonulin, fecal calprotectin

Structured N-of-1 protocols evaluating the top-ranked candidates (NAC, metformin, sub-antimicrobial doxycycline) with pre-specified biomarker endpoints would represent the logical next step in translating this screen to clinical practice. Several of these drugs have well-established safety profiles including pediatric use, reducing the ethical and practical barriers to N-of-1 investigation.

4.6 Limitations

This analysis has several important limitations that must be acknowledged.

No clinical validation. None of the drugs identified in this screen have been clinically tested for Marshall syndrome. All therapeutic hypotheses are based on mechanism-of-action extrapolation from related biological contexts (disc degeneration, osteoarthritis, complement disorders, senescence biology). The gap between mechanistic rationale and clinical efficacy is frequently large, and many mechanistically promising drug repurposing candidates fail in clinical testing.

Mechanism extrapolation. The cascade model itself is an interpretive framework synthesized from published research across multiple fields. While each individual cascade is supported by peer-reviewed literature, the integration of all eight into a single interconnected network, and the specific connectivity relationships between cascades, have not been experimentally validated as a unified system in COL11A1-mutant tissue.

Tissue-specific pharmacokinetics. Many of the target tissues (intervertebral disc, articular cartilage, vitreous humor) are avascular or semi-avascular compartments with limited drug penetration. Systemic drug concentrations may not achieve therapeutic levels in these tissues. This limitation is particularly relevant for oral supplements and medications that must traverse multiple biological barriers to reach the disc or vitreous.

Drug-drug interactions. The combination strategy proposes concurrent use of multiple compounds whose interactions have not been systematically studied in combination. Unexpected pharmacokinetic or pharmacodynamic interactions could alter efficacy or safety profiles.

Pediatric considerations. Marshall syndrome is a congenital condition, and many affected patients are children. Several candidates (rapamycin, statins, dasatinib) have significant pediatric contraindications or limited pediatric safety data. The ranking methodology accounts

for this, but the overall applicability of the screen to the pediatric population requires additional scrutiny.

Ranking subjectivity. The composite ranking score, while structured, involves subjective weight assignments and evidence quality assessments. Different weighting schemes would produce different rankings. The ranking should be interpreted as a prioritization framework rather than a definitive ordering.

Publication bias. The literature search underlying this screen is subject to the standard limitations of publication bias—positive results are more likely to be published than negative results, potentially inflating the apparent evidence base for some candidates.

4.7 Future Directions

The following priorities emerge from this analysis:

1. **N-of-1 trials.** Structured N-of-1 protocols for the top 3 candidates (NAC, metformin, sub-antimicrobial doxycycline) with pre-specified biomarker endpoints, conducted under physician supervision in willing Marshall syndrome patients.
2. **Cx43 hemichannel pharmacology.** Development of selective, safe Cx43 hemichannel inhibitors that do not affect cardiac gap junctions. Tonabersat represents the most advanced candidate and warrants investigation in musculoskeletal contexts.
3. **In vitro validation.** Testing of top-ranked candidates in COL11A1-mutant chondrocyte or nucleus pulposus cell cultures (patient-derived fibroblasts reprogrammed to relevant cell types) to assess cascade-specific efficacy.
4. **Biomarker panel development.** Standardization of a multi-cascade biomarker panel that can serve as a composite endpoint for clinical trials in type XI collagenopathies.
5. **Allele-specific gene silencing.** Design and validation of COL11A1-specific ASOs or CRISPR guide RNAs, leveraging the proven approaches from sister collagen disorders [45, 46].

5. Declarations

Competing Interests

The authors declare no competing interests. No pharmaceutical company, supplement manufacturer, or commercial entity was involved in the design, conduct, analysis, or reporting of this screen.

Funding

This work received no external funding. It was conducted entirely as a patient-family-driven research synthesis. The authors have no financial relationships with any entity that could have influenced this work.

Data Availability

All data underlying this analysis are derived from publicly available peer-reviewed publications and public databases (PubMed, ClinicalTrials.gov, DrugBank). The complete drug screen data, including all candidates not shown in the main tables, are available on the Drug Repurposing Screen (drug-repurposing-screen.html) page of the Marshall Syndrome Resource. The cascade model is described in full at the Unified Model (../basics/unified-model.html) page.

Ethics Statement

This study involved no human subjects, animal subjects, or clinical data. It is a literature-based synthesis and does not require ethics committee approval.

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This manuscript has NOT been peer-reviewed. It is a preprint-format synthesis prepared by a patient family for informational purposes. It does not constitute medical advice. The authors are not medical professionals. No drug identified in this screen has been clinically tested for Marshall syndrome. All treatment decisions must be made with qualified physicians who understand the individual patient’s complete medical history, current medications, and risk factors. Drug interactions between agents listed here have not been systematically studied in combination. Nothing in this manuscript should be acted upon without physician guidance.

← Drug Repurposing Screen (./drug-repurposing-screen.html)

Research Roadmap → (./research-roadmap.html)